

1 Milestone 2: KPI Engineering and Dashboard Planning

1.1 Introduction

The second milestone of the MEDTRACK DV project focuses on the identification, calculation, validation, and presentation of Key Performance Indicators (KPIs) for hospital performance and operational analytics.

KPIs are measurable values used to evaluate hospital operations, patient flow, resource utilization, and patient care outcomes. They help transform the prepared datasets into meaningful performance metrics that support data-driven decision-making.

The KPIs were calculated using the cleaned and standardized analytical datasets developed during Milestone 1. The validated metrics were subsequently used in Tableau to design KPI cards and interactive dashboards.

1.2 Objectives of KPI Development

The major objectives of this milestone are:

- To identify relevant performance indicators for hospital operations and patient care.
- To calculate hospital admission, length of stay, readmission, and bed utilization metrics.
- To validate the calculated KPI values against the analytical datasets.
- To present the metrics through clear and understandable Tableau visualizations.
- To establish a quantitative foundation for hospital performance monitoring.

1.3 KPI Identification and Calculation

The following KPIs were identified and calculated for the MEDTRACK DV project.

1.3.1 Total Admissions

Definition:

Total Admissions represents the total number of patient admission records available in the hospital overview dataset.

Calculation:

$$TotalAdmissions = COUNT(PatientID)$$

Validated Value:

10,000

This KPI provides an overview of the total patient admission volume represented in the analytical dataset.

1.3.2 Average Length of Stay (LOS)

Definition:

Average Length of Stay measures the average duration patients spend in the hospital.

Calculation:

$$AverageLOS = \frac{TotalLengthofStay}{TotalNumberofPatients}$$

Validated Value:

15.51days

This indicator supports the analysis of patient stay duration and hospital operational efficiency.

1.3.3 Bed Occupancy Rate

Definition:

Bed Occupancy Rate measures the proportion of hospital bed capacity occupied during the period represented by the dataset.

Calculation:

$$BedOccupancyRate = \frac{OccupiedBeds}{TotalBeds} \times 100$$

Validated Values:

- Total Beds: 2,991,110
- Occupied Beds: 1,990,850
- Occupancy Rate: 66.56%

The calculated occupancy rate provides an indication of hospital bed utilization and available capacity.

1.3.4 Readmission Rate

Definition:

Readmission Rate represents the percentage of eligible patient records classified as readmitted in the readmission dataset.

Calculation:

$$ReadmissionRate = \frac{NumberOfReadmittedPatients}{TotalEligiblePatients} \times 100$$

Validated Values:

- Total Eligible Patients: 10,000
- Readmitted Patients: 4,142
- Readmission Rate: 41.42%

This KPI is used to analyze readmission patterns and support further investigation of patient care outcomes.

1.3.5 Department Efficiency

Definition:

Department Efficiency is a composite indicator intended to compare hospital departments using multiple operational and patient care factors.

The planned weighting scheme is:

Performance Factor	Weight
Patient Satisfaction	30%
Waiting Time	25%
Length of Stay	20%
Readmission Rate	25%
Total	100%

The weighted approach combines multiple performance factors into a single comparative measure.

The previously calculated summary values were:

- Average Department Efficiency: 52.03
- Minimum Score: 1.29
- Maximum Score: 99.40

Data validation note: The department analytics dataset used for the Tableau dashboards did not contain the calculated department efficiency score. Therefore, the score requires inclusion and validation in the final analytical dataset before it can be presented as a directly reproducible Tableau KPI.

1.4 KPI Validation

KPI validation was performed to ensure that the calculated metrics were consistent with the available analytical datasets.

The following validation activities were carried out:

- Verified the total number of admission records in the hospital overview dataset.
- Checked the average length of stay against the prepared patient records.

- Compared occupied beds and total bed capacity to calculate the occupancy percentage.
- Verified the number of readmitted patients and the total eligible records.
- Reviewed the department efficiency summary and identified the missing efficiency score field in the department analytics dataset.

A separate bed utilization calculation produced an occupancy rate of 66.67%, using 1,969,490 occupied beds and 2,953,930 available beds.

This differs from the 66.56% occupancy rate calculated using the hospital overview figures. The two calculations use different underlying values and require reconciliation before a single occupancy KPI is finalized.

1.5 KPI Implementation in Tableau

The validated KPI values were used as the basis for developing KPI cards in Tableau.

The KPI cards were designed to display important hospital performance metrics in a concise and understandable format.

The implementation activities included:

- Creating individual KPI worksheets for important hospital performance metrics.
- Displaying KPI names and calculated values using text marks.
- Formatting KPI values with appropriate number formats, font sizes, and visual emphasis.
- Organizing KPI worksheets within the Hospital Overview dashboard.
- Preparing the calculated metrics for integration with the remaining project dashboards.

1.6 Milestone 2 Outcome

Milestone 2 established the KPI framework for the MEDTRACK DV project. The major hospital performance indicators were identified, calculated, and reviewed using the prepared analytical datasets.

The KPI calculations provide a quantitative foundation for the Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization dashboards.

The milestone also identified data consistency issues related to bed occupancy calculations and the missing department efficiency score field. These issues must be addressed to ensure consistency and reproducibility in the final dashboard implementation.

Table 1.2: Summary of Hospital Performance KPIs

Total Admissions	10,000	Patients
Average Length of Stay	15.51	Days
Bed Occupancy Rate	66.56	Percentage
Readmission Rate	41.42	Percentage
Average Department Efficiency	52.03	Score